GLP LABORATÓRIOS

FOOD SAFETY

# Radiometric analysis in food

Safety, compliance and traceability for food products in the domestic and international market.

- Committed to the future
- Brazilian laboratory
- Broadest scope of radiometric analyses



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02 · CONTEXT

# Why quantify radionuclides in food

GLP Laboratórios performs the radiometric analyses required by the domestic and international market, in addition to issuing reports for assessing radioactivity levels in accordance with the requirements of importing countries.

01 **Export requirement**
Each market defines its own set of radionuclides and limits.

02 **Consumer safety**
Food safety control through validated radiometric methods.

03 **Certification and audit**
Traceable and auditable reports for quality programs.



# Main segments

Radiometric analyses applied to different food matrices.



### Meat products

Beef, pork and poultry



### Dairy

Milk and dairy products



### Grains and cereals

Coffee, yeasts, nuts, peanuts



### Beverages

Non-alcoholic and mineral water



### Fruits and vegetables

Fresh and processed products

# Required radionuclide tests

Examples of radionuclides required for food products originating from Brazil.

| REQUESTING BODY | MATRIX | REQUIRED RADIONUCLIDES |
|---|---|---|
| **FDA**<br>Food and Drug Administration | Food in general | I-131 · Cs-134 + Cs-137 · Ru-106 + Ru-103 · Sr-90 · Pu-238 + Pu-239 + Am-241 |
| **FAO**<br>Food and Agriculture Organization | Food in general | Pu-238 + Pu-239 + Pu-240 + Am-241 · Sr-90 + Ru-106 + I-129 + I-131 + U-235 · S-35 + Co-60 + Sr-89 + Ru-103 + Cs-134 + Cs-137 + Ce-144 + Ir-192 · H-3 + C-14 + Tc-99 |
| **European Union** | Infant food, dairy products and liquids | Cs-134 + Cs-137 · Sr-90 · I-131 · Pu-239 + Am-241 |
| **Portaria 146** | Food in general | Cs-134 and Cs-137 · I-131 · Sr-90 |

# Main determinations

Radiometric determinations organized by radionuclide families.

### Cesium and Iodine

Cs-134 | Cs-137 | I-131

### Radium and Lead

Ra-226 | Ra-228 | Pb-210 | Pb-214

### Thorium and Uranium

Th-232 | Th-230 | U-238 | U-235 | U-234

### Strontium and Ruthenium

Sr-90 | Sr-89 | Ru-103 | Ru-106

### Plutonium and Americium

Pu-238 | Pu-239 | Pu-240 | Am-241

### Other radionuclides

K-40 | Ac-228 | H-3 | Co-60 | Ir-192

# How we get to the result

Four complementary techniques cover alpha, beta and gamma emitters in any food matrix.

### Alpha Spectrometry

Radiochemical separation, radionuclide concentration and silicon detection.

### Gamma Spectrometry

Cesium, iodine, potassium and other natural and anthropogenic gamma emitters.

### Liquid Scintillation

Low-energy beta radionuclides, such as Tritium.

### Proportional Counter

Total alpha and beta and, after separation, Sr-90, Ra-226, Ra-228 and Pb-210.

# Qualifications and differentiators

Credentials and commitments that underpin analytical reliability.

### ABNT NBR ISO/IEC 17025

Technical competence in laboratory testing · CRL 1266

### ANSN

National Nuclear Safety Authority · licensed facility

### IAEA · ALMERA Network

International proficiency testing in environmental radioactivity

### Committed to the future

Work focused on safety, quality and responsibility.

### Brazilian laboratory

National structure for specialized technical support.

### Broadest radiometric scope

Breadth of determinations for different requirements.

# From sample to report

Five stages between first contact and the certificate ready for shipment.

1

### Scope definition

Destination market, matrix and required radionuclides.

2

### Sample shipment

Guidance on proper collection, quantity and packaging.

3

### Preparation and analysis

Analytical chemistry and application of the technique suited to the matrix.

4

### Quantification

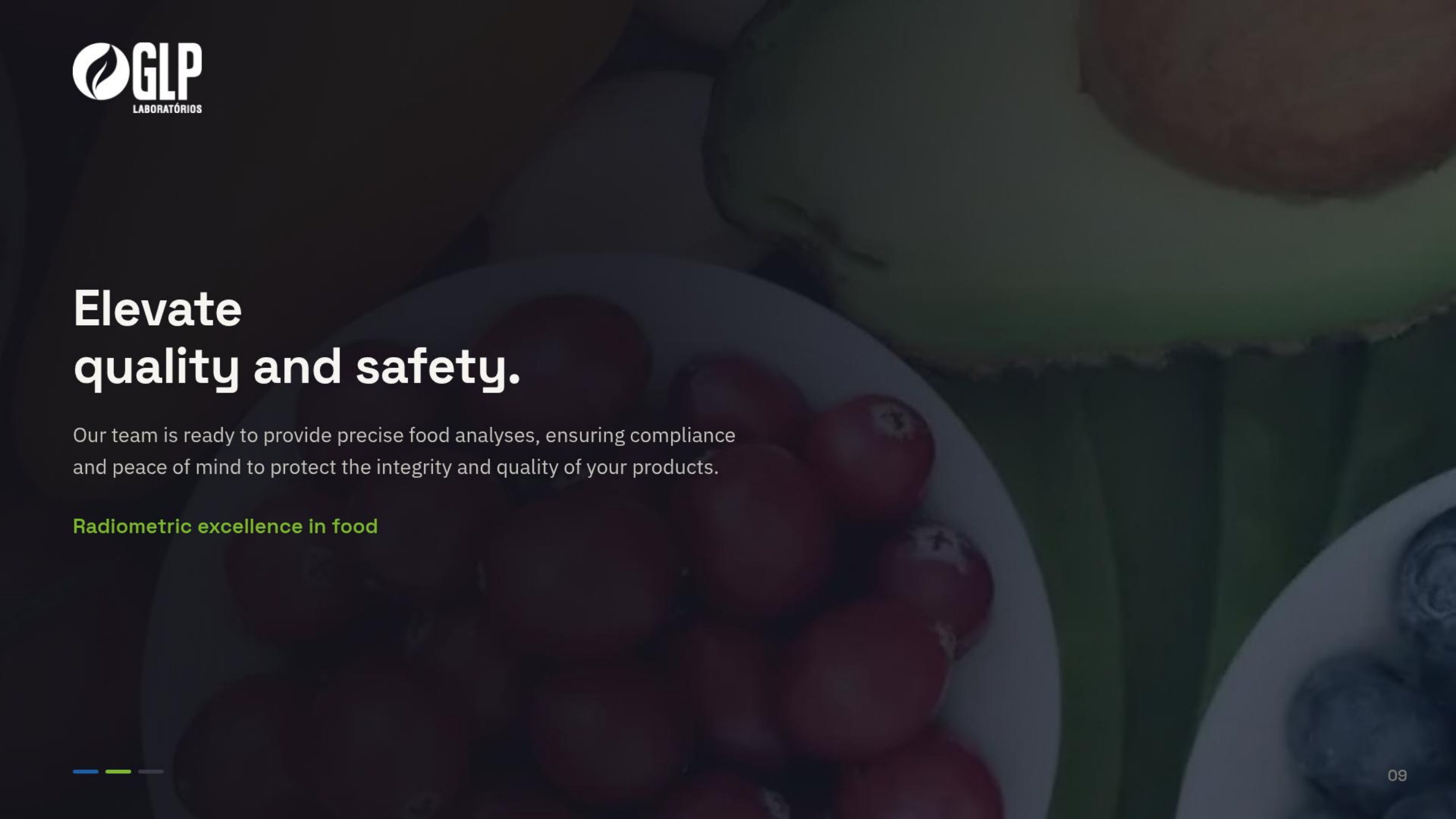
Activity concentration with internal quality control.

5

### Certificate of analysis for export

Traceable and auditable radiometric certificate of analysis.

Reduced turnaround times and tracking through the **Online Results** portal.



# Elevate quality and safety.

Our team is ready to provide precise food analyses, ensuring compliance and peace of mind to protect the integrity and quality of your products.

**Radiometric excellence in food**

GLP LABORATÓRIOS

# Get in touch.


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**Brazil’s largest radiometric laboratory.**